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The Alexandroff Present and Minkowski Spacetime: Why it Cannot Do What it has Been Asked to Do[1]

In recent times there have been interesting attempts to introduce a *mind-independent* present in Minkowski spacetime, a structure that has *also* been regarded as capable of explaining some aspects of our subjective experience of time, in particular the fact that we share a "now" but not a "here".[2] Considering that Minkowski spacetime is the arena for three out of the four interactions postulated by contemporary physics, this claim, if correct, would be remarkable. Against the prevailing opinion of last century's major physicists and philosophers, we would have in fact discovered that it is possible to find a now (even a "transient one") in any contemporary physical theory whose spatiotemporal arena has the structure of Minkowski spacetime. At the same time, we would have gone some way toward a rapprochement of "the manifest image"[3] of time − characterized by a cosmically extended, transient now − with the physical image, traditionally dominated by the picture of a block universe in which "the present is absent" because regarded as purely mind-dependent.

The main thesis of this paper is that in Minkowski spacetime the so-called "Alexandroff present" (henceforth, for short, "Alex"), introduced by Arthur and

1 Thanks to Richard Arthur and Steven Savitt for extensive discussions and written exchanges over many of the topics discussed here. From them, I have in any case learnt a lot also in the past. Our exchange has reduced, but possibly not completely eliminated, the number of misunderstandings on my part. Carl Hoefer has read a previous version of this paper and helped with critical comments and questions.

2 Richard Arthur, "Minkowski spacetime and Dimensions of the Present", in: Dennis Dieks (Ed.), *The Ontology of Spacetime*, Amsterdam: Elsevier, 2006, pp.129-155; Steven Savitt, "The Transient nows", in: Wayne C. Myrvold, Joy Christian (Eds.), *Quantum Reality, Relativistic Causality, and Closing the Epistemic Circle,* The Western Ontario Series in Philosophy of Science 74, Amsterdam: Springer 2009, pp. 339-352.

3 This famous expression is in Wilfrid Sellars, "Philosophy and the Scientific Image of Man", chapter 1 in Wilfrid Sellars, *Science, Perception, and Reality*, New York: Humanities Press, 1963 pp. 40, where he contrasts the manifest image of the world with its scientific image.

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Savitt[4] is neither a physically relevant property[5] nor a very good *explanans* of our temporal experience. Therefore, it should be dropped.

The paper is structured as follows. In 1 I will present Savitt's two main motivations for having a present in Minkowski spacetime, and the corresponding two senses in which physics might be said to need an objective now. In 2 I will introduce the main features of Alex, and the alleged link between the agreed-upon temporally extended character of the present of our experience (specious present) and its supposed "interactive nature". In 3 I will discuss what I refer to as "Alex's *dilemma*" between its marginal role in physics and its unfaithfulness to our experience. In 4 I will illustrate the former horn of the dilemma, by presenting some possible uses of Alex in mathematical physics. These will be shown to be insufficient to conclude that Alex has an important role in physical theories. In 5, I will illustrate the latter horn of the dilemma, by defending at the same time an alternative, better explanation of the spatial extendedness of our experienced nows, based on empirical data concerning our psychophysical threshold for regarding two light signals as being temporally successive.

1 Savitt's two motivations for having a present in Minkowski spacetime

The *first*, and possibly more important, motivation consists in showing that the task necessary to close "Shimony's circle" between the objective aspect of time and its felt, subjective aspect, can, *at least in principle*, be completed:

> The program [of closing the circle] envisages the identification of the knowing subject (or, more generally, the experiencing subject) with a natural system that interacts with other natural systems.[6]

I am very sympathetic toward this project,[7] as it is one of the inspiring Muses of the philosophy of time.

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4 Alex was originally suggested by Howard Stein, "On Relativity Theory and the Openness of the future", in: *Philosophy of Science* 58, 1991, pp. 147-167, and elaborated upon by Wayne Myrvold, "Relativistic Quantum Becoming", in: *The British Journal for Philosophy of Science*, 54, 2003, pp. 475-500. These authors' claims, however, were less ambitious than Arthur's and Savitt's.

5 Except, possibly, in recent attempts at building a mathematical setting in which causality can be studied independently of geometrical and differentiable structure Keye Martin, Prakash Panangaden, "Spacetime topology from causality", in: arXiv:gr-qc/0407093v1, 2004.

6 Abner Shimony, "Reality, causality, and closing the circle", in *Search for a Naturalistic World View*, Vol. I. Cambridge: Cambridge University Press, 1993, p.40.

7 Arthur does *not* share this motivation with Savitt in an explicit way. Of course, this does not amount to saying that he sees no point in trying to explain objectively why

However, Savitt's *second* motivation for introducing an objective now in Minkowski spacetime is, I take it, in potential *conflict* with his ambition of "closing Shimony's circle". This second motivation consists in his attempt to show that Einstein's often quoted "worry" that physics cannot grasp "the Now" is, in some sense, unfounded:[8]

> Once Einstein said that the problem of the now worried him seriously. He explained that the experience of the Now means something special for man …That this experience cannot be grasped by science seemed to him a matter of painful but inevitable resignation (Carnap 1963, 37, my emphasis).

Parallel to Savitt's two motivations, there are *two* senses in which one could think that physics needs an objective now. According to the first, current physics is to be regarded as *incomplete* because it has not yet captured an allegedly objective property "*being now*", a property that − unlike "being massive" or "being charged" − has so far escaped all physical theories. While this first sense might be regarded as being too strong, I take it that it is the *only* sense in which one would be entitled to conclude that "there is a viable alternative" to Einstein's worry (see note 8).

The second sense in which physics might need a now is much *weaker*, as it amounts to claiming that a physically objective now − possibly corresponding to some objective spatiotemporal structure − is needed *either* to explain some key features of our subjective experience *or* in the practical application of physical theories by human beings, *or* to give a precise semantic interpretation of our common use of "now" regarded as an indexical.

Given that Arthur and Savitt agree with me and others that "now" is an indexical, and that indexicals cannot appear in laws, it seems safe to conclude that the only reason to introduce an objective present in Minkowski spacetime is linked to the *second* sense, the one that Arthur and Savitt are really after. As we will see, however, even in this weaker sense, Alex turns out to be an *arbitrary choice*, since other spatiotemporally extended structures, or other non-geometrical explanations, can do the explanatory job in a more satisfactory way.

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time is experienced in the way we do (personal communication).

8 After quoting from physicists and philosophers holding that there cannot be a now in physical theories, Savitt writes: "What I hope to do in this paper is show that there is a viable alternative picture to these views, a picture that includes, in some sense, a *now* and the passage of time" (ibid., p.340). This second motivation, unlike the first, is shared also by Arthur: "Indeed it is false to say that physics does not take the "now" into account" (ibid., p. 151, n. 35).

2 Alex's features in relation to the extended character of our experience

It is now time to look at the features of "Alex" in some more detail, since they are crucial in order to assess its suitability for explaining our experience of time, or for accounting for some pragmatic uses of the "present" in cosmology, as when we claim that, for example, "galaxies in our universe formed a certain number of billions year *ago*", or that "*presently* the universe is expanding".[9]

The Alexandroff's present *relative* to a segment of any timelike curve delimited by points *a* and *b* is the intersection of the future light-cone whose vertex is the beginning event *a* with the past light cone whose vertex is the end event *b* (as customary, a temporal orientation is assumed). The Alexandroff's present *relative to the segment a-b of a worldline* − call such a set Alex [*a,b*] − is then by definition the set of points in the above mentioned intersection. The (relational) objectivity of Alex [*a,b*] depends on the invariance of the relation of *causal connectibility* within the special theory of relativity. In fact, it turns out that any event in the set Alex [*a,b*] *is both a possible effect and a possible cause of events on the segment of the worldline*, and for this reason it could be regarded as the set of events with which my body or any other physical system modelled by a worldline can in principle *interact* during the time-like interval a,b. Notice for example that in fig. 1 below, point *c*, which is outside Alex [a,b] but inside event's *b* past light cone, can be a possible cause of many events on the segment, but is not in *mutual* causal contact with all of it, since it cannot, in its turn, be caused by events belonging to the segment.[10] More precisely, Alex [*a*,*b*] is the set of *all* points *z* such that there exist two points *x* and *y*, belonging to the segment *a*, *b* for which we have *both* C_p*xz and* C_p*z*y, where C_p is the relation of past causal connectibility.[11]

One more point worth noting is that the length of the timelike interval under consideration may vary from context to context, and this is certainly an advantage of this conception: as we know from ordinary discourse, "now" can pick out my present perception, the present historical moment, or even the "present stage" of cosmic expansion.

9 Arthur, ibid., p. 152.

10 Event a and event c are not causally connectible, and the "first" event that could cause c is below a, or "earlier than a", see fig. 1.

11 This more rigorous rendering is due to Myrvold, ibid. p. 480. Stein talked about Alex as the set of events *contemporaneous* with a given segment of a worldline, Stein, ibid, p. 159.

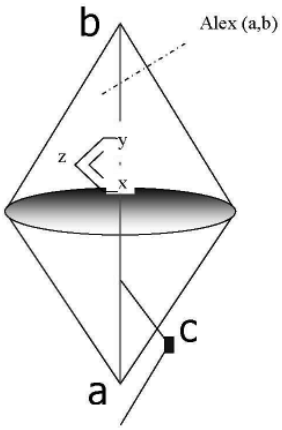


FIG. 1

Imagine that the segment *a*, *b* represents, with due idealizations, my point-like body *looking* at my room "right now". Why should my present experience be represented as temporally extended along the segment *a*-*b*? There are two possible arguments in favour of such a hypothesis. One comes from phenomenological evidence stemming already from James, Husserl and other psychologists and phenomenologists, linked to studies related, say, to our perception of music or language. The wealth of experimental evidence coming from this corner and militating *against* a point-like present is quite strong.

A second, more speculative argument has been put forth by Stein. On Stein's hypothesis, the temporal extendedness of the present of our experience is due to the conditions that he regards as necessary for a conscious experience to occur. These conditions require *coordination*, i.e., several mutual interactions of the parts of the brains with themselves and with the external environment. Notoriously, in relativistic physics, interactions take time, and it is this *empirical hypothesis* that could explain, among other things, why our experience of the present is not knife-edged, as James put it, but "specious", or temporally extended.

In a word, the idea of a spatiotemporally extended present like Alex's − a set of events that is in mutual possible causal interaction with our bodies and brains experiencing something − in Stein's opinion would be a consequence of the fact that the stability of the various parts of our brains and of the external physical objects around us needs several coordinations, which are realized only thanks to several mutual physical influence.[12]

Agreed: one immediate advantage of choosing Alex as a model of our experienced present is that it allows *a prima facie* plausible explanation of the reason why we all share the same now,[13] as well as of our "natural" intuition − belonging

12 Stein, ibid. p. 161.

13 For a different explanation, see Jeremy Butterfield, "Seeing the Present", in: *Mind* 93,

to "the manifest image of time − that the now, or the present moment, *appears* to us as being cosmically extended: Alex is temporally thin and spatially fat.[14] The "illusion" of a cosmically extended now, notoriously, had to be unmasked by Einstein in order to lay the foundations of the special theory of relativity. Since any perception ascertaining the simultaneity of two events is always *local*, the attribution of a relation of simultaneity to two *distant* events necessarily entails some conventional element.

Considering the fact that our experience of time is always local, it is also natural to claim that *if* we want to represent our experience of time in Minkowski spacetime, we must rely (on segments of worldlines measured by) *proper* time, and not on *coordinate* time. Dieks has pointed out, correctly in my opinion, that we don't need global nows to accommodate our *local* experience, since different spatial hyperplanes intersecting the same point or the same short timelike segment − an idealized representation of our body − *lead to the same temporal experience*.[15]

However, notice that Alex's spatial extendedness, while sufficient to explain the corresponding aspect of our experienced now, is by far *not* sufficient to regard Alex as a good *explanans* in the second sense specified above. Different extended regions of spacetime might also succeed in explaining this aspect of our experience, and Savitt must somehow show that Alex is the best explanation we can give, a difficult task that, as we are about to see, he tries to undertake without success..

Summarizing my presentation of Arthur and Savitt's view of Alex, I agree with them on the following two crucial points: (1) the present of our experience *looks* extended; (2) however, the notion of time on which to base the representation of the experienced present in Minkowski spacetime is *proper* time rather than *coordinate* time.

1984, pp. 161-76 and Craig Callender, "The Common Now", in *Philosophical Issues* 18, 2008, pp. 339-361.

14 For this reason, Alex has also been used to explain the origin of the belief in presentism. "… presentists … are inclined to accord a kind of ontological privilege to that with which they can (then) interact. What they can interact with, at a particular near-momentary subregion R of their worldtube, is the contents of that region's Stein Present. But … such a Stein Present has very little temporal thickness and is very large in spatial extent. Such a region is easily mistaken for an instantaneous, global present.", in Ian Gibson and Oliver Pooley "Relativistic Persistence," in John Hawthorne (Ed.), *Philosophical Perspectives*, Vol. 20, *Metaphysics*. Oxford: Blackwell, 2006, p.169. For an attack on the significance of the presentism/eternalism debate in Minkowski spacetime, see Steven Savitt, "Presentism and eternalism in perspective", in D. Dieks (ed.), *The Ontology of Spacetime*, Amsterdam: Elsevier, 2006, pp. 111-127, Yuval Dolev, "How to square a non-Localized present with special relativity", in D. Dieks (ed.), *The Ontology of Spacetime*, Amsterdam: Elsevier, 2006, pp and 177-190, and Mauro Dorato, "The irrelevance of the presentist eternalist debate in Minkowski spacetime", in D. Dieks (ed.), *The Ontology of Spacetime*, Amsterdam: Elsevier, 2006, pp. 93-109.

15 Dennis Dieks, "Becoming, relativity, and locality" in D. Dieks (Ed.), *The Ontology of Spacetime,* 1, Amsterdam, Elsevier, 2006, pp.157-175.

3 Alex's oscillation between its uselessness in physics and its unfaithfulness to our experience

The dilemma that I want to present in this section is a consequence of a tension between Savitt's two motivations for introducing Alex in Minkowski spacetime, which for brevity I will refer to as "Shimony's circle" and "Einstein's worry".

(1) The undisputable objectivity of Alex − needed by Savitt and Arthur to overcome *in some sense* Einstein's worry − makes it unsuitable to account for, or explain, the subjective present;

(2) Any success in explaining the subjective present with a geometric structure of Minkowski spacetime would make a solution to Einstein's worry highly implausible.

In the rest of the paper, I will argue in favor of these two claims.

As explicitly recognized by Arthur (and, I think, implicitly also by Savitt) Alex, *qua* objective spatiotemporal structure, is *not* a model for the subjective present, that is, it is not "the set of all those events of which we are consciously aware at the moment of considering them".[16]

One of the reasons for this claim has been put forward by Arthur himself: the set of events that we can actually be conscious of during an interval of time is always a *subset* of the set of those events that are in *actual* mutual communication with our bodies during that interval. This is not simply due to our selective attention − filtering out from our awareness many events that *are* in our relative Alex − but also to the fact that Alex is the set of *possible* and not of *actual* causes and effects affecting our bodies during a specious present. Therefore, many events in Alex will not be registered by our consciousness, and Alex cannot coincide with the events represented in our subjective present.

For analogous reasons, however, the set of events in Alex is not picked out by our everyday uses of "now", at least not on the hypothesis that the latter is regarded (as Arthur, Savitt and I think it should) as an indexical. While the context-dependence of such uses is well accounted for by the variable length of the time-like curve with extremes a,b, and while it must be admitted that our uses of the indexical "now" need not necessarily refer to the set of events that we subjectively regard as present, the *intentions* of the speakers are certainly relevant to fix the set of events picked out by the indexicals we use. And by no stretch of imagination could we claim that, by uttering "now", we typically intend to refer to all the events with which we are in possible mutual causal communication (namely, to Alex). On Savitt's and Arthur's part, this could be stipulated only with a remarkable degree of arbitrariness, since in our usage of "now", the idea of intending to refer (token reflexively) to events that are *simultaneous* with our speech acts seems much more prominent. It does not seem very plausible to claim that we can be wrong about what we intend to pick out when we use the word "now", even though it must be admitted that Alex does contain simultaneous events.

16 Arthur, ibid., p.151.

However, if Alex is neither (nor is intended to be by its proponents) the subjective present, nor can it be the set of events that we typically intend to pick out when we use the temporal indexical "now", why should we believe that − simply in virtue of its spatial fatness − it is a good explanation of our "experience" of speakers and of conscious agents (assuming, of course, that the spacetime we live is Minkowski)? Granting that Alex can have *some* explanatory virtue is not sufficient to remove an impression of arbitrariness. To avoid this problem, Arthur and Savitt should show that Alex is, if not the best available explanation of our experience, a very plausible candidate.

As an argument against this plausibility, , notice that when we look, say, at the starry sky, we believe that the light we see is part of our present experience. However, the events corresponding to the emission of light from, say, a supernova, occurred various years before our perception, and are therefore certainly *not* in our *momentary* specious present, which typically lasts more than half a second but less than a few seconds. It then follows that the events on the supernova that are responsible for our perception are outside Alex's, *and yet they are pre-theoretically regarded by us as present*. The fact that Alex has not been proposed as a physical model of our experienced present (it does not even contain events that *we* regard as present) weakens its explanatory force: any strong *explanans* should at least describe correctly its *explanandum*.

As an additional argument in favour of this weakness, notice that the example of the supernova entails lack of supervenience of our mental experience of time over Alex, at least if we suppose that [*a*,*b*]'s temporal length is less than a few seconds.

In fact, a difference in what we regard as *psychologically* present (say, we first look at the moon and then we immediately see a supernova in the sky), by including events outside Alex, is not reflected in a difference of the subvenient, physical base, since Alex remains the same in the two cases (i.e., immediately before and at the moment in which we see the supernova). But if our subjective experience does not supervene on Alex, it cannot provide a strong explanation of our temporal experience, at least to the extent that the existence of a relation of supervenience of a family of mental properties B over a family of physical properties A is a *necessary* (albeit not sufficient) condition for any explanation of B by using A. And even though neither Arthur nor Savitt have proposed Alex as a possible physical basis of supervenience for our mental experience, I think that they will grant that, as a consequence, the explanation that it can provide cannot be but *weak and partial*. It could be replied that whenever we are looking at closer stars, say, the Sun, we could *stretch* the segment [a,b] that represents our present into a much longer temporal intervals (16 minutes), so as to include the original electromagnetic emission from our star. Such an "elastic" procedure could possibly be iterated even when we look at the nocturnal sky with a powerful telescope. Wouldn't this be a way to rescue Alex from the charge of being "unfaithful" to our experienced present?

After all, we have already noticed how the length of the present is pragmatically dependent on the events we are referring to.

Suppose we accept this "elasticity" as a necessary consequence of the variability of the contexts of utterance of the temporal indexicals. This "stretchy Alex", however, would be parasitic on a completely *different model of the experienced present*. The amount of the stretch would in fact come to depend on the distance of the events we are looking at, and therefore on our regarding as present the set of events on the past light-cone centered in our bodies. This would be tantamount to identify the present with the events *on* the past light-cone. But then why not preferring the past-light cone to Alex as a possible explanation of the extended character of our experience? If the length of Alex comes to depend on the distance of the events which we are interacting with in perception, one must accept that Alex is less successful an explainer of our temporal experience than the past light cone.[17]

Arthur considers the possibility of representing the present of our experience with the set of events lying on the past light cone, but complains that such a move would be equivalent to eliminating "any distinction between past and present".[18] This is doubtlessly correct. He should have realized, however, that such an elimination is the *essential* feature of our experience of the present. In fact, we are completely unaware of the time lag due to the speed of light, and unconsciously attribute light an infinite speed. The approximation, for all practical purposes of our earthbound life, is extremely good, since objects that are close "enough", as noted by Butterfield, typically don't change much after light bounces off them and reaches our retinas.[19]

I think that what we are facing here is a deep, acute conflict between the descriptive needs of physics and those of the psychology/phenomenology of the experience of time. On the one hand, if we adopt Einstein's standard convention of simultaneity in terms of "worldline-orthogonality", from the viewpoint of physics the events of the emission of the electromagnetic radiation from the stars or the Sun must be regarded as being in the causal past of our perceiving bodies. On the other hand, if we regard the emission of radiation from celestial objects and our act of perceiving it as *simultaneous* with our perceptions, we are much more faithful to our experience, but we eliminate any difference between past and present.

We could of course adopt a criterion of simultaneity that is *different* from the one that is customarily adopted, and yet much closer to our experience of time: according to this different method, we could regard as simultaneous with our perception any event on the past light cone. However, as already noted by Einstein, this method of fixing simultaneity would be quite impractical for physical needs, because it would make simultaneity depend on the particular spatiotemporal position occupied by the clock:

17 See last section for more arguments.

18 Ibid., p. 151.

19 Butterfield, ibid.

We might, of course, content ourselves with time values determined by an observer stationed together with the watch at the origin of the co-ordinates, and co-ordinating the corresponding positions of the hands with light signals, given out by every event to be timed, and reaching him through empty space. But this co-ordination has the disadvantage that it is not independent of the standpoint of the observer with the watch or clock, as we know from experience.[20]

The impracticality of the method of simultaneity discussed in this quotation shows that, even if we decided to modify Einstein standard criterion of simultaneity, the descriptive needs of physics and those of psychology would diverge.

The *second* argument against the possibility of stretching indefinitely Alex according to need is this: if the extension/duration of the now is totally context-dependent, i.e., dependent on what we observe, its *physical* meaning would seem to become proportionally weaker. Alex would still be objective, but characterized by a length that is somehow mind or observer dependent!! So once again our attempt to solve Einstein's worry fails to help solving the issue raised by "Shimony's circle".

These conclusions are generalizable to structures other than Alex and the past light-cone above. Once we accept (for physical reasons) Einstein's standard criterion of simultaneity, it follows that even if we *could* successfully model the psychological present by using some other structure, like the past light-cone discussed above, we would end up with something which introduces some deep discrepancies with the constraints of the physical description of the world. And the simplest hypothesis that explains this gap is this: our natural folk-physical belief about what is present is to some extent simply illusory, since it *falsely* assumes the presentness (*simultaneity* with our perception) of whatever falls into our visual fields, supernova explosions included.

Einstein expressed the illusory nature of our experience of time quite clearly:

The illusion which prevailed prior to the enunciation of the theory of relativity … had its origin in the fact that in our everyday experience we can neglect the time of propagation of light. We are accustomed on this account to fail to differentiate between "simultaneously seen" and "simultaneously happening"; and, as a result, the difference between time and local time is blurred.[21]

While in Savitt's intentions Alex is doubtlessly no more than *a first step* in the attempt to explain our subjective temporal experience − so that he would agree with me that much more needs to be done − my attempt here is to show that it is a step in a *wrong direction*.

20 Albert Einstein et. al., *The Principle of Relativity*, transl. by W. Perrerr and G.B. Jeffery, Dover Publication, New York, 1905/1952, p. 39.

21 Albert Einstein, "Physik und Realität", *Journal of The Franklin Institute* 221, pp. 313-347, English translation in *Ideas and Opinion,* New York, Bonanza, 1936, p. 299.

In fact:

1) In virtue of its objectivity, Alex cannot fulfil the aim of closing Shimony's circle. Savitt's central question of bridging the gap between the time of physics and the time of the manifest image is still up for grabs, and must be studied by other means, namely neurophysiology and psychology,[22] and not by combining some arbitrary physical structure with some features of our experience.

2) Attempts to close Shimony's circle by selecting structures out of Minkowski spacetime are destined to be unsuitable for the descriptive purpose of physics, and therefore for proving Einstein's worry unfounded.

Against Savitt, I think we can conclude that his two motivations cannot be had at the same time, and one of them must give way. In the next two sections, we will see that, as a matter of fact, the situation is even worse: Alex can serve *neither* purpose. The physical irrelevance of the interactive present on the one hand (4), and further arguments in favour of the unfaithfulness of the interactive present on the other (5), will complete my objections to the introduction of Alex in a physical theory whose spatiotemporal arena is Minkowski's.

4 The physical irrelevance of Alex

It would be unfair to claim that the Alex has *no role whatsoever* in physical theories. For instance, the condition of "strong causality" in a Riemannian manifold is provably equivalent to the condition that the Alexandroff topology coincides with the manifold topology, or to the condition that the Alexandroff topology is Hausdorff.[23] The basis of this topology is given by the intersection of the set of future directed timelike curves originating from p with the set of past directed timelike curves originating from q, with p earlier than q: $\{I^+(p) \cap I^-(q) : p, q \in M\}$.

Readers will recognize Alex's present relative to the segment $[p, q]$ of the timelike curve, except that the Alex also contains lightlike geodesics, and is therefore $\{J^+(p) \cap J^-(q) : p, q \in M\}$, where $J^+(p)$ is the set of future directed causal curves originating from p and $J^-(q)$ is the set of past directed causal curves originating from q.

We need to distinguish an important fact separating Minkowski spacetime from general relativistic spacetimes, and investing the possibility of deriving metrical facts from topological/causal facts alone. While in Minkowski spacetime the manifold topology coincides with the Alexandroff topology, in order to obtain the

22 In any case, Savitt cannot be accused of trying to replace empirical research by philosophical reflection.

23 Martin and Panangaden, ibid. The Alexandroff topology "is the coarsest topology in which all the interiors of light cones are open". See Lawrence Sklar, *Philosophy and Spacetime Physics*, University of California Press, Berkeley, 1985, p. 81. More generally, an Alexandroff topological space is one in which the intersection of *any* family of open sets (also non finite) is still an open set.

desired equivalence in general relativity, we need to impose the further condition of *strong causality*, which is equivalent to the fact that the manifold has *no* "almost-closed causal curves", or no curves that come arbitrarily close to intersecting themselves. This means that "for every point p in the manifold M and for every neighbourhood O of p, there is another neighbourhood V of p contained in O such that no causal curve intersects V more than once".[24]

In Minkowski spacetime it can be shown (following Robb) that the postulation of a "before-ness" relation, plus some axioms that such a relation must satisfy, is sufficient to fix the fact that an event is in the interior of a past or of a future light cone.[25] This means that all the topological properties of regions of Minkowski spacetime can be recovered just by looking at the relation of causal connectibility, and the topology of spacetime *can* be the Alexandroff topology.

In a word, the main foundational interest of the Alexandroff topology lies in the possibility of determining metrical facts from topological facts involving the interiors of the light cones of Minkowski spacetime. The possibility of quasi-closed curves in general relativistic models, however, divorces the manifold topology from the Alexandroff topology, because in spacetime with almost closed causal curves not all causal automorphisms can also be homeomorphisms.

However, it should be admitted that the illustrated role of the Alexandroff topology in foundational studies is a far cry from claiming that Alex is *relevant* for physics. In particular, there is no connection between the Alexandroff topology and temporal properties, despite the fact that *stably* causal spacetimes − for which the above relationship between manifold topology and Alexandroff topology a fortiori holds − do have a cosmic time.[26]

The latter concept introduces another important issue: Arthur claims that it is in the *application* of physics, rather than with respect to physical laws, that the now has a role. He correctly points out that it makes a great deal of difference with respect to available evidence whether it is supposed that the Big Bang occurred 14 or 26 billions *years ago*: "the relativity of the now to certain events (humans having theories) does not detract from its objectivity".[27] However, here the question is not the objectivity but the relevance of the now for physics. In the sentence just quoted, we can give the truth conditions of the above tensed expression by simply pointing out that, relatively to a certain class of events in spacetime, where we are tenselessly located at the moment in which we use a certain theory, the Big Bang

24 Robert Wald, *General Relativity*, Chicago University Press, Chicago, 1984, p.196. Strong causality is weaker than stable causality, which in turn is necessary and sufficient for the existence of a cosmic time function.

25 Sklar, ibid, p.79. See also John Winnie, "The causal theory of spacetime", in: J. Earman C. Glymour, and J. Stachel (Eds.), *Foundations of spacetime theories*, Minnesota University Press, Minneapolis, 1977, pp. 134-205.

26 The use of closed Alex diamonds in algebraic quantum field theory as a basis to define an algebra of observables is also interesting but here it cannot be expanded upon.

27 Ibid, p.152.

has occurred 14 billions years before those events, rather than 26 billions before. But let us even grant a role to the tensed expression "billions of years ago" (we can accept that indexicals have essential meanings), and let us suppose that the "located" above is "located now". Why should this expression entail reference to the set of events that is in mutual causal interaction with the worldline that, initiating with the Big Bang, intersects the areas of spacetime in which we now refer to that first event?

Given the possibility of using a cosmic time, as it is customary in Friedman-Robertson-Walker's cosmological models, we can interpret that tensed language (billions years ago) as referring − in a token reflexive way, and in the moment in which we use the theory − to the proper time of the "fundamental observer" associated with the mean motion of matter in our local cluster or supercluster of galaxies. Such a proper time can then be extended, given certain assumptions of isotropy, to the proper time of other "fundamental observers".

Note that (i) this standard manoeuvre is fully compatible with a denial of the possibility of using cosmic time to defend presentism or a metaphysics of tenses, but also that (ii) our need of referring to the beginning of time and space cannot be said to provide evidence for the postulation of Alex. In virtue of the Cosmological Principle in fact, "having the age of ... years" is a property that the Universe has in *all* spacelike-related regions *now* sharing the same values of pressure and density that we are measuring. The property in question is therefore not relative just to us users of the physical theory in a particular point/region along a particular worldline, as seems to be implied by a structure like Alex.

5 Another model of the present of our experience?

The point of this final section is to reinforce the thesis already argued for that not only is Alex unsuitable to represent or explain some features of the present of our experience, but also that no other structure of Minkowski spacetime could do the job. This will allow me to conclude that we should not try to use such a spacetime to explain or account for features of the present of our experience, in the weakest possible sense of explain or account. Given the facts (already argued for) that:

(i) Alex has no important applications in *physical* theories;
(ii) it does not seem a plausible, strong and non-arbitrary explanation of the extendedness of our subjective present, and
(iii) It does not correctly pick out the events we intend to pick out when we use "now" in ordinary language
(iv) these seem the only reasons to introduce it,.

I conclude that we should drop it.

With respect to (ii), it seems reasonable to add to the previously discussed counterexamples that Alex should also contain events that we regard as present

simply because their temporal separation is so small as to be inferior to our capacity to tell them apart. There is a lot of experimental evidence in this field, which for our purpose we will consider only synthetically, and which, however, is very relevant in order to explain the central feature of our experience of the now, namely its spatially extended appearance.[28]

Owing to the reaction time that is necessary to distinguish *two* light signals as temporally *successive*, we perceive as *simultaneous* events that are separated by less than a given threshold. In general, the threshold necessary to tell two signals apart is different from person to person, and varies also from sensorial modality to sensorial modality. But in average, and for *visual stimula*, empirical research indicates that there is a threshold of 30ms for a person to be able to tell *two* flashes of light apart, let alone determine their temporal order, for which at least 45 ms are necessary. Multiplying the threshold of 30ms by the speed of light *c*, we get 30 x 10^{-3} x 300.000 km/h = 9000 km. A flash *a*, that originated 30 ms on the surface of a sphere whose radius is 9000 km, but simultaneous with event *b* where I am located, registers on my retina 30ms later, at event *d*.

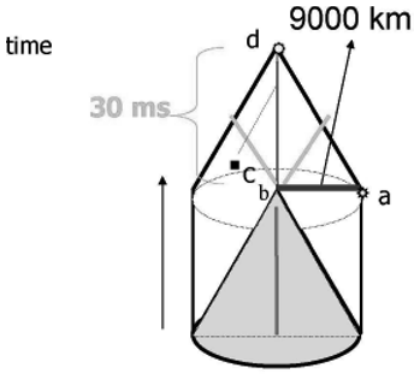


FIG.2

More in general, we perceive anything inside a sphere centered in *b*, and whose radius is 9000km, as simultaneous with all the points between *b* and *d*, which are separated by 30ms. This sphere, which is really a cone, can be visualized correctly in the shaded area of fig.2. By remodelling Savitt's suggestion in the right way, *the succession of those cones could give us the transient now*. Should we then propose the grey area in the figure as a more faithful replacement of Alex and claim that the succession of the grey, conical regions along worldlines represents the passage of time?

I would be ready to maintain that the grey wedge or conical model is much less arbitrary than Arthur and Savitt's proposal, as it is based on well-confirmed

28 Some of this literature is covered in Callender (ibid.), who correctly invites philosophers of time that, like ourselves, are trying to bridge the experiential with the physical time, to devote more attention to experimental findings in neuropsychology.

psycho-physiological data, and that, therefore, it describes more precisely the extended character of our psychological present, in which perceived space, regarded as Leibniz's order of coexistence, is a construction of our brains. Note that since the light signals intersecting one after the other the vertical segment b-d are *physically* temporally separated but cannot be *perceived* as being temporally separated, they should be treated as part of the set of points that are subjectively regarded as present. Events like *c* are *not* in Alex [*b*,*d*] but are perceived as simultaneous with *b* (at the center of the sphere) and with *a*, at its surface, and therefore *are* in the same psychological present of the observer located in *b*. Events like these count as other instances of events that Alex does not contain, and that we are forced by our innate perceptual machinery to regard as present.

6 Conclusion

In sum, we should not continue the game that Arthur and Savitt initiated by playing it with a different (and more appropriate) ball, namely the conical model.

We have already noted how the fusion of the past with the present, of what is far away with what is occurring right here, explains why our experience of time "misrepresents" the physical, objective temporal separation of events. The misrepresentation, or better, the temporal fusion of timelike-related or light-like related events, is due to the fact that the time of occurrence of events is merged with the time of our perceiving them. To acknowledge that the present is extended means to acknowledge that the "past-there" is fused with the "present-here": our treating them as simultaneous is due to the psychophysical threshold needed to tell events apart, plus the considerable speed of light. From an evolutionary viewpoint, this fusion does not represent any terrible distortion of our environment, insofar as objects, as noted by Butterfield (see note 19), do not change their properties too much during the short time needed by light to inform us about them and by our nervous system to process the relative information.

Notice that if we acknowledge that our experience of time is a fusion of events that from the viewpoint of physics are temporally separated, then it does not make much sense to try to represent with a geometrical/physical structure the set of events that *we* represent as being present, except by specifying that these events are part of a single experience of simultaneity.

Consequently, we *do* need to distinguish the "physical present" of events and the psychological present. The former, due to the relativity of simultaneity, in Minkowski spacetime must be represented as pointlike (each physical event is present, from its own "perspective", when and where it occurs). It is only the latter that is extended, and can include many pointlike physical events, many of which are *temporally separated from the other*: violations of achronality are admissible only for the psychological present, but not for the physical present.

In a word, if we want to bridge the gap between the time of physics and the time of our experience, as Savitt correctly urges us to do, it is important to begin our work by recognizing their important differences.

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